



Towards a sociotechnical ecology of artificial intelligence: power, accountability, and governance in a global context

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Abstract

Contemporary artificial intelligence (AI) technologies, particularly those based on foundation models and released at scale, are globally entangled and made up of a complex array of interrelated actors, practices, and transnational flows of resources. The rapid pace at which AI systems are being developed and distributed, is driving significant societal and planetary transformations. While much of the international agenda around governing AI has converged around downstream matters of safe deployment and use, deeper systemic issues—including power concentration, uneven environmental costs, or the asymmetric extraction of data and labour by technology companies—remain contested and unresolved areas of debate. In this paper we centre these systemic challenges, and locate governance entry points aimed at fostering more just futures. We conceptualise contemporary large-scale AI as a sociotechnical ecology comprised of interrelated actors, practices, and asymmetrical resource flows. Using the lens of *infrastructural inversion* within social studies of infrastructure, we trace the actors involved in the making of AI technologies, their interdependences, and long-term infrastructural continuities that shape them. We argue that new AI models and systems are not unprecedented but are instead built upon and shaped by pre-existing infrastructures, entrenched market relations, and socio-historical patterns. By making visible the sites of accountabilities and technical and non-technical intervention in the AI ecology, we identify four governance imperatives for sustainable and equitable AI governance: (1) Decentralising AI infrastructure, (2) Advancing environmental and epistemic justice through pluriversal AI governance, (3) Instituting cross-border data and data work governance, and (4) Enhancing international coordination, participation and solidarity.

Keywords Political economy of AI · Global AI governance · Epistemic injustice · Environmental injustice · Infrastructural inversion

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1 Introduction

The last few years have seen major developments in Artificial Intelligence (AI) applications and systems and with them the recognition of potential for socially transformative effects, both positive and negative. While great promise has accompanied the emerging capabilities of state-of-the-art AI systems in various sectors and domains, a large and growing body of literature points to a range of systemic and planetary-scale risks and negative impacts of these technologies [1–3]. The global race to develop ever more powerful AI systems has thus simultaneously triggered a global sense of urgency to develop policy mechanisms and governance approaches that are sufficient to address such risks and harms [4]. A highly contested arena of competing interests surrounding AI innovation and the development

of national and international AI regulation has accordingly gained prominence.

Foundation models¹ are the latest computational techniques to power a new generation of AI, where large amounts of data are used to generate more sophisticated outputs compared to predecessor technologies. These models use deep learning/neural network architectures in a self-supervised manner, which enables the models to be trained on vastly greater quantities of data. Foundation models are trained on “broad,” web-scale data sets, including a range of data types such as images, text, audio data, and time-related data, which are mostly scraped from the internet and can be applied to a wide range of tasks and purposes [5, 6]. Because of this, they require significant material, energy, human, and natural resources to produce. The techniques involved are not new, but the scale of the data used for training, the coordination of global networks of labour to process these massive amounts of data, the increased architecture complexity, and the development of increasingly powerful processors and other computational resources have together made possible new levels of predictive and generative sophistication.

Prevalent narratives of AI innovation tend to conjure up a deterministic and sequential process where technological advancements are increasingly faster and exponential [7, 8]. The hype surrounding AI has sparked a goldrush among Big Tech companies, AI start-ups, and venture capitalists that has enabled foundation models to be in a state of continuous evolution to improve their capabilities and performance. Well-funded AI labs are on a path of continual iteration and growth, as is evident with the rapid development, public release, and announcement of new models like OpenAI’s GPT 5, Anthropic’s Claude Sonnet 4, xAI’s Grok 3, DeepSeek R1, and so on. Bolstered by the implicit inevitability of model growth and hyperscaling, these advancements are accompanied by promises of incremental progress toward an impending horizon of “Artificial General Intelligence” (AGI) [9–11]. While highly contested [12], these promises have driven much of the discourse on AI safety and governance.

The hefty infrastructure, human labour, and data demands of these AI technologies and, in particular, generative AI (GenAI) mean it can only be developed by a handful of

well-funded technology companies around the world such as Anthropic, Cohere, Meta, OpenAI, Microsoft, and Alphabet – the latter two being substantial funders of other firms.² Academic institutions and government funders are making major investments but are challenged to match those made by the largest technology companies, leaving most high-level AI innovation in their hands. This concentration of research and development investment into private ventures raises concerns that the benefits of most cutting-edge AI technology – are becoming vastly centralised in a small group of powerful actors, rather than being more broadly shared in society [13].

Due to their unspecified general utility, unprecedented energy and material requirements, significant labour demands, expense, and globally distributed nature, GenAI technologies present new challenges for law, policy, practice, the biosphere, and collective well-being [2, 14]. To face this “problem of many hands” [15], it has become increasingly urgent to interrogate how governance approaches can grapple with the layered and global entanglements of AI. Indeed, a great deal of the discourse and policy intervention on AI risks and harms has focused on tackling issues arising downstream. These include a host of ethical and technical issues of bias, privacy, and copyright violations, psychological harm, fraud and misinformation, reliability, accuracy and security, among others [16, 17]. On the other hand, the upstream harms of these technologies and the associated systemic issues of power concentration, exacerbation of inequalities, contributions to ecological degradation, and fragmentation of the AI regulatory landscape, appear to remain largely unaddressed, or are cast as too complex and intractable in the current governance landscape [18–20].

In this paper we foreground these systemic issues with the aim of identifying key sites of decision-making and thereby opportunities for equitable and sustainable governance. We ask: What are the critical governance considerations arising at each point of AI production, dissemination, and integration? And what interventions are needed to address the current social and environmental issues posed by AI?

Drawing insights from existing analyses of the political economy of AI and Science and Technology Studies (STS), we delineate what we term the *sociotechnical ecology of AI* as an analytical device to chart a complex array of actors, practices, processes, and cross-border flows of natural and material resources, capital, data, labour, and knowledge. We use the analytical approach of *infrastructural inversion*

¹ The term foundation models, coined by researchers at Stanford, alludes to them forming the base architecture of other systems. For example, ChatGPT and the conversational features of the Bing search engine are applications built on top of successive versions of OpenAI’s GPT foundation model. Most foundation models are Large Language Models (LLMs) that are primarily directed towards the interpretation or production of content that uses human languages, including spoken/written language and programming code. A key distinctive feature of foundation models is their multi-modality. They can be configured to process many types of data, including images and sound in addition to text. Lack of coherent terminology is an issue within the AI community with other terms like “frontier models” referring to similar systems having emerged more recently.

² By example, Microsoft reportedly invested over \$13 billion in OpenAI. Other companies, including Google and Salesforce have invested hundreds of millions of dollars each in companies and their own internal development of foundation models since late 2022. See <https://www.technology.org/2023/03/24/ai-spring-four-takeaways-from-major-releases-in-foundation-models/>.

developed in social studies of infrastructures to identify the constitutive elements, interdependencies, and longitudinal continuities between key actors in the AI landscape with the aim of locating potential sites of accountability, power, and agency. This lens has been productively used to foreground the hidden relational dimensions of large technical systems. It enables us to “invert” the default view of systems in the wild, to look into their making, materialities and historical trajectories. Our contribution is primarily conceptual but with the actionable aim of locating the potential levers of intervention throughout the making and global deployment of AI which could engender more just and sustainable futures. We ground our analysis on a specific subfield of AI, namely foundation models—given their distinctive character of being globally entangled, large scale, and general-purpose. In doing so, we recognise that some of the governance issues highlighted here are connected to, or overlap with, those of other narrower AI techniques and of data-driven systems at large [21, 22].

The paper is structured as follows:

We begin (in Sect. 2) by exploring the current landscape of AI governance, paying special attention to the pressing systemic and global challenges that have arisen with foundation models and GenAI. In Sect. 3, we introduce infrastructural inversion as a critical approach to conceptualise AI’s global embeddedness and address such systemic and global governance challenges. We then empirically trace (in Sect. 4) the sociotechnical ecology of AI, identifying key practices, actors, and their interdependencies as well as potential sites for governance intervention. We conclude by proposing an agenda of four governance imperatives for engendering more sustainable and equitable AI technologies.

2 The landscape of systemic and global AI governance challenges

Given the globally embedded nature of AI, governing the technology has been increasingly framed as a cross-border issue calling for the development of laws, standards, and policies for regulating AI at a global level [23]. This framing emphasises in particular the role of international organisations and multistakeholder cooperation in addressing AI’s global challenges [21] and in formulating governance mechanisms driven by societal needs and human rights considerations [4, 24, 25]. However, cross-border concerns are not addressed solely through global-level interventions. There remains significant continued interest in the design of technical tools and policies aimed at aligning AI technologies with ethical values like fairness, accountability and transparency at the level of states and individual organisations

[26–29]. Considering the full breadth of potential intervention points, we can broadly define AI governance as *the establishment of policies, practices, norms, institutions, and incentives to steer the production and future trajectory of AI systems toward a desired goal*.

AI governance efforts to date encompass a range of responses to the wide-ranging legal, ethical, and moral concerns arising from contemporary AI systems. A number of national, regional, and global initiatives have emerged as attempts to grapple with such concerns [30–34]. There is no singular approach to identifying, mitigating, or addressing the risks and harms associated with AI while steering innovation in socially beneficial directions. AI governance efforts are continuously being revisited and adapted to both advancements of AI technologies and changing socio-political contexts [30, 35]. Because of this, the field of AI governance remains highly contested and dynamic.

A growing body of research has shown that the current wave of large-scale AI innovation has the potential to replicate and dramatically amplify risks and harms to individuals, society, and the biosphere [2, 36]. While landmark measures such as the European Union Artificial Intelligence Act (EU AI Act) –the first regulation in the world to adopt a risk-based approach to AI—explicitly acknowledges systemic risks and harms, the majority of governance activity today remains concentrated on downstream safety and security issues. This trend is reflected in the proliferation of national and regional AI safety and security institutes, whose efforts focus largely on self-regulation, industry standards, and technical measures like red teaming, model evaluation, and safeguards against toxic or discriminatory outputs [16, 37].

In this paper we wish to draw closer attention to systemic and geopolitical challenges pertaining to the globally distributed nature of the AI value chain, and the need for actionable upstream and systemic responses. Building on our engagement with emerging AI policy debates and a growing literature on global disparities and the human rights risks of AI, we focus on some of the most pressing yet largely unaddressed global challenges: the concentration of market and geopolitical power by dominant AI firms; the inequitable extraction of personal data, creative works, and labour for the training of large AI models; the unevenly distributed environmental impacts of AI; and the global asymmetries in the AI regulatory landscape.

2.1 Power concentration

The costs of developing and deploying AI, which have only increased with the proliferation of GenAI, entail access to large-scale data, technical expertise, and great compute capacity that can only be afforded by a handful of well-funded globally operating firms [38]. These entities are

further identified as principal developers and intellectual property (IP) owners of the most advanced AI technologies, including having oversight of fundamental research, infrastructure, production, and international distribution. The vast financial costs constitute high barriers to entry for smaller competitors [1] facilitating market consolidation and corporate capture of wider regulatory systems [13, 39, 40]. Through their control of the core compute infrastructure of large-scale AI, private technology firms have entrenched their power by gatekeeping downstream AI innovation and its trajectory of travel [41, 42]. This results in throttling prospects of meaningful competition and further limits the possibility of impactful policy interventions from public or independent bodies.

Indeed, governments worldwide are aligning with the extractive financial models forwarded by large technology firms and have established a dependency on industry rather than implementing regulations that rein in the formation of corporate monopolies [43]. In response to these emerging bottlenecks across the production of large-scale AI, there has been an increased focus by competition authorities worldwide to examine AI-affected markets [44, 45].

In addition, the significant concentration of technoscientific production in highly industrialised countries like the US, EU nations, and China results in strategic influence and geopolitical asymmetries, impacting which countries get to shape the trajectory of AI innovation. Such competition among industrial nation blocs is observed through different dynamics like the disruption of global AI markets created by the emergence of the Chinese AI model DeepSeek R1 and the European Union's (EU) prioritisation of both in situ innovation and recent announcement to invest €200 billion of investment in AI [46]. DeepSeek R1's open-source approach has accelerated global competition by signalling alternative more cost-effective paths to AI development, threatening the dominance of Silicon Valley, and lowering barriers to entry for smaller players. These developments underscore how geopolitical competition in AI innovation increasingly shapes broader international power relations, as nations grapple with maintaining technological leadership amidst rapidly evolving global competition.

2.2 Cross-border data, information, and labour extraction

In March 2023, Italy ordered the immediate block of ChatGPT until OpenAI set measures to comply with the EU's General Data Protection Regulation (GDPR). Soon after, France, Germany, and Spain opened investigations into ChatGPT, and the European Data Protection Board launched a dedicated task force to coordinate possible enforcement actions between EU data protection authorities. Similarly,

the Ibero-American Network of Data Protection Authorities initiated a coordinated action to supervise ChatGPT services and its compliance with data protection law in their jurisdictions. And both the Colombian data protection authority and the United States Federal Trade Commission launched investigations into the compliance of OpenAI with their respective domestic laws. More recently, with the release of DeepSeek R1, Italy's data protection authority took a similar action to that in response to ChatGPT and launched an investigation into the Chinese company.

This series of government scrutinies signals a regulatory reaction to the inherently extractive nature of contemporary large AI models. AI has become a tool for global surveillance and a threat to fundamental rights, such as the rights of consent, transparency, and purpose limitation, amongst others. The AI companies that are developing foundation models, do not typically obtain consent from data subjects or establish a legitimate interest for using their data. As it is standard practice for large AI model producers to reveal little about the nature and sources of data they use for training, there are limits to what data subjects are likely to know, which is therefore limiting on their practical assertion of rights. As a result, data protection authorities have called into question the lawfulness of the data processing practices to train GenAI [47–49].

Existing regulations are an imperfect and incomplete solution to the multiple forms of extraction of large scale AI. Data protection laws lack extraterritorial force and most of AI producers are located in countries—such as the US and China—which are outside of the jurisdiction of the more stringent EU data protection laws. While data subjects in covered countries are entitled to legal protection from any data controller regardless of national origin, the practicality of international enforcement is limited and tempered by the lobbying power of big technology companies. Some types of AI, such as foundation models, have the potential to infringe copyright and IP law due to their memorisation capabilities, the non-consensual scraping of data, and failure to provide a legal basis for the legitimate use of public domain and copyrighted content. The EU addresses the legality of data mining of copyright works in the training datasets through its Digital Single Market Directive, which allows the reproduction or extraction of copyright works by research organisations and cultural institutions with lawful access. It also allows the use of lawfully accessible works whose copyright owners did not express any prohibition to their work being used for data mining. In addition, the EU AI Act mandates general purpose AI provide transparency about the training data and policy on how to respect copyright law. Yet, whether and how these mandates could have a bearing beyond the EU remains an open question.

AI companies' race to collect large-scale datasets to build complex AI models has also exacerbated the growth of the global labour market involved in the development and curation of such datasets [50]. AI firms often outsource these tasks to Global South countries with weaker labour laws and much lower labour costs, which enables them to distance themselves from the treatment of the workers, often characterised by precarious and exploitative conditions [51–54]. Workers contracted by outsourcing companies are often bound by non-disclosure agreements, which limit their rights to freedom of expression [55]. These arrangements contribute to making the conditions of data workers opaque as well as creating an asymmetry of accountability that protects platforms and AI companies while holding data workers responsible for errors and data quality [50].

Regulatory responses to these exploitative practices in the global AI labour supply chain therefore remain an urgent challenge. Whilst the EU AI Act [32] includes provisions related to value chain governance, it excludes policy concerns associated with the impacts of data development processes on the rights and wellbeing of data workers. Likewise, the EU's Directive for Platform Workers [56] does not address these concerns. The European Corporate Sustainability Due Diligence Directive [57] makes small progress in this realm and mandates companies to put in place measures to assess negative impacts on human rights of their activities across their supply chain, where it includes 'social cost of poorly paid microwork' [55]. In the US, while the National Institute of Standards and Technology (NIST) Framework identifies risks across the AI value chain, similar to the EU AI Act, it does not include provisions for data workers protection [58]. As Khan & Hanna [50] note, regulatory instruments that include references to datasets focus largely on downstream impacts rather than those arising at the development stage of the datasets and, therefore, renders invisible and leaves governance gaps in the activities where data workers are most affected.

Despite the global asymmetries in the AI value chain, the Global South has made significant contributions to AI development and research [59]. Indian software developers have been among the leading contributors to AI projects on GitHub, while China was the global leader in AI-related research publications [60, 61]. Countries in East Asia and the Pacific have increasingly dominated the global landscape of AI granted patents, rising from 75.2% to 82.4% in 2023. In contrast, North America's share declined from 21.21% to 14.23% [61, 62]. Researchers in Africa, Asia, and the Middle East are leading the development of language models tailored to local languages, cultural contexts, and digital sovereignty [63]. Big tech firms have not only been sourcing talent from the Global South for high-skilled roles [64], but have also increasingly invested in establishing

tech hubs and research labs in countries like India, Brazil, and Kenya [65–68]. There is a growing presence of Global South professionals across the broader technology sector, including those working abroad, remotely, or through outsourced roles.

However, as Steinhoff [69] warns, this expansion of high-skill roles does not necessarily translate into equitable returns or working conditions. Steinhoff argues that Big Tech's outsourcing practices often transform previously privileged data science roles into precarious, underpaid forms of ghost work, particularly for workers based in the Global South. While these professionals may occupy high-status positions in name, their autonomy and bargaining power are increasingly undermined by centralised control mechanisms, such as proprietary training platforms (e.g., AWS Educate, Salesforce Trailhead) and corporate-sponsored capacity-building initiatives. In response to these dynamics, national AI strategies in countries like Canada and Brazil have begun to emphasise domestic capability-building, often through partnerships with big tech and reforms in education and workforce development.

2.3 Unequal environmental impacts of AI

One of the most pressing governance challenges pertaining to large-scale, general-purpose AI is the considerable entanglements of these technologies with socio-environmental injustices. For example, energy intensive AI systems have both significant direct and indirect negative environmental impacts, particularly in the course of model training and in the associated operation of large-scale data centers [70]. They can involve the generation of terawatt hours of electricity, some using fossil fuel combustion resulting in poor air quality and downstream pollution, and others hazardously seeking alternative energy sources such as nuclear to meet escalating power demands. Similarly, data centers require water resources for cooling and result in massive volumes of fresh water consumption, and the exponential need for rapid hardware manufacturing, replacement, and disposal result in e-waste landfill and hazardous materials [71]. These present challenges are exacerbated by limited approaches to accurately measuring compounded effects and by the lack of transparency by tech companies which rarely release data on their carbon footprint [3, 72, 73]. Various tools, such as the Machine Learning Emissions Calculator [74], Carbon-tracker [75], Green Algorithms [76], and Code Carbon [77] have been developed as attempts to evaluate carbon emissions and the environmental impact costs of developing and deploying foundation models and GenAI applications. Yet despite these efforts, Kneese [78] warns that they are frequently co-opted by private technology companies using forms of corporate greenwashing. Global South countries

in Africa, Latin America and Asia, for instance, remain disadvantaged by socio-ecological injustices through ongoing and unsustainable extractive economics and resource driven conflicts exacerbated by the unrelenting reliance of AI and data infrastructures on critical minerals and natural resource consumption [79].

Governance bodies and mechanisms at the international level responding to these glaring issues include the International Organisation for Standardisation (ISO) and its published environmental management system and related standards (e.g. 14001; 14040) [80] and the International Telecommunication Union's (ITU) Recommendation L.1410 [81], which employs a life cycle assessment (LCA) methodology. LCA is a methodology that entails moving beyond the deployment phase to encompass the social, material, and ecological impacts of the global logistical infrastructures that enable the production of AI-intensive solutions. These impacts include the extraction of rare earth minerals, chip manufacturing, hardware transportation, and the accumulation of e-waste following decommissioning of an AI system. Additional developments at various levels have emerged that seek to implement ways of analysing, tracking, and managing environmental costs of AI, including recent proposals for a centralised repository for emissions reporting [82], the HULK energy efficiency benchmark tool [83], and a lifecycle analysis of emissions from manufacturing to deployment.

However operationalisation of end-to-end governance mechanisms, such as the LCA, heavily rely on agreed multi-stakeholder frameworks for measuring, tracking, and reporting environmental indicators across the AI lifecycle. Reaching consensus on these metrics is critical in enacting comprehensive governance measures that seek to proactively address and mitigate the adverse environmental impacts within the emergence of wider AI regulatory frameworks. To date, regulatory efforts to address environmental concerns have nonetheless been limited and fragmented. Article 40 of the EU AI Act requires providers to comply with yet to be developed EU standards on reporting and documentation of the processes to improve AI systems' resource performance. These include the consumption of energy and environmental resources by high-risk AI systems and the energy-efficient development of general-purpose AI. Despite the European Commission's proposal being considered the "first comprehensive attempt in the world to regulate AI from a fundamental rights perspective" [84], the EU AI Act has been criticised for lacking sufficiently binding requirements in addressing environmental impacts and in reducing the carbon footprint in line with net zero goals [85]. Moreover, an evaluation produced by the Council of Europe's Ad hoc Committee on Artificial Intelligence

(CAHAI) openly notes that considerations of sustainability appear to be "significantly underrepresented" [86].

2.4 Regulatory fragmentation and asymmetries

The global AI policy and regulatory landscape has accelerated in response to the rapid distribution of GenAI.³ For instance, the Association of Southeast Asian Nations published an expanded ASEAN Guide on AI Governance and Ethics, which supplements the 2024 version of the guide that excluded generative AI policy recommendations [87]. China has published the Interim Measures for the Management of Generative Artificial Intelligence Services [88] which not only applies to domestic companies but also to overseas GenAI providers with activities in the domestic market. And the final version of the EU AI Act was delayed in order to include specific obligations for general-purpose AI model providers, including technical documentation, EU copyright law compliance, and disclosure of detailed summaries of content used for training. Various regulatory philosophies are leading to different approaches in regulating AI development and use at the international level.

Regulation prioritisation, for instance, can differ among stakeholders in emphasising systemic risk mitigation, or clear requirements, given that prioritising innovation is a driver for many governmental and industry actors. Countries vary in their regulatory prioritisation; however, even among risk weighted regulatory approaches, these can differ in how they are considered. NIST's AI Risk Management Framework in the US comprises non-binding recommended actions [89], whereas the EU AI Act is a binding legislative instrument, and when fully enacted will directly regulate the use and application of AI, as is Brazil's rights-based and risk-based Bill 2338/2023, and the Council of Europe's Framework Convention on AI and human rights, democracy, and the rule of law for the use of AI systems for adopting public authorities and private actors acting on their behalf. In addition, in some countries, initiatives to advance hybrid frameworks combining binding legislative instruments with adaptive and multistakeholder compliance mechanisms are taking place. For instance, Brazil's data protection authority (ANPD) officially announced the AI Regulatory Sandbox in 2025 through which AI projects will be evaluated in relation to personal data processing, societal impacts, and other criteria, to promote responsible experimentation with AI models [90].

³ According to the Digital Policy Alert Activity Tracker, the number of AI policy or regulatory changes has been steadily increasing over the years, with significant jumps occurring following the rise and widespread adoption of foundation models and GenAI: 33 AI policy or regulatory changes in 2020, 79 in 2021, 94 in 2022, 336 in 2023, and 538 in 2024. In just the first two months of 2025, 158 policy or regulatory changes have already been recorded.

Some scholars warn that the divergence in AI regulatory developments emerging in different jurisdictions can lead to a fragmented AI regulation landscape. The situation has the potential to undermine international cooperation on AI governance and exacerbate the challenges for regulatory interoperability and addressing shared risks [21, 91–93].

Notwithstanding diverging national regulatory philosophies, multi-state agreements around AI regulatory concerns have continued to be viewed as a priority. The AI Safety Summit in the UK in November 2023 aimed to achieve global consensus in areas of regulatory concern around risks related to frontier AI. Twenty-eight governments, including the US, the UK, China, six EU member states, Brazil, Nigeria, Japan, and Saudi Arabia and leading AI companies signed the Bletchley Declaration affirming that AI should be designed, developed, deployed, and used in a manner that is safe, human-centric, trustworthy, and responsible.⁴ The most recent international AI policy gathering of this kind was held in France in February 2025 and was rebranded the “AI Action Summit,” signalling a broadening of concerns beyond the initial focus on AI risks and safety. Despite this, the Summit ultimately played to the tune of multinational technology companies pushing for deregulation [94]. The diverging and competing aims of the countries represented at the Summit was also reflected in both the US and the UK refusing to sign the Summit’s Statement on Inclusive and Sustainable Artificial Intelligence for People and Planet [95]. The BRICS+ countries have also joined efforts to agree on commitments on AI. With a critical view on how power is distributed among AI governance actors, after the Rio de Janeiro 2025 meeting, they issued a declaration emphasising the need for global AI governance to operate both under national regulatory frameworks and the UN Charter, whilst respecting sovereignty and security [96].

Intergovernmental organisations are advancing broader instruments, such as the UN’s Global Digital Compact [97] that reinforces the need for ensuring that digital technologies (including AI) are used responsibly and for the benefit of a global community, whilst addressing uneven AI adoption globally and fostering a safe and inclusive digital environment. In March 2024, the UN General Assembly adopted by consensus a US-led resolution on “Seizing the opportunities of safe, secure and trustworthy artificial intelligence systems for sustainable development” supported by more than 120 Member States [98]. This global resolution continues in a similar vein to the Bletchley Declaration [99] in addressing the risks associated with irresponsible or malicious applications of AI systems that could detrimentally impact human rights and fundamental freedoms.

⁴ See <https://www.gov.uk/government/publications/ai-safety-summit-2023-the-bletchley-declaration/the-bletchley-declaration-by-countries-attending-the-ai-safety-summit-1-2-november-2023>.

UNESCO’s Recommendation on the Ethics of Artificial Intelligence [34] has served as a model for regional efforts, as exemplified by the Santiago Declaration to promote ethical AI development across Latin America and the Caribbean [100] and the Windhoek Statement on Artificial Intelligence in Southern Africa [101]. Similarly, the regional AI governance landscape in Africa had significant advancements, including the African Union’s African Digital Compact [102] and the Continental AI Strategy [103].

It is clear that the current landscape of global AI initiatives has a disadvantaged representation of the global majority [17, 104]. According to the UN’s AI Advisory Body, 118 countries do not participate in some of the most influential international AI governance initiatives, with more than 75 per cent of countries in Africa, Asia Pacific, and Latin America and the Caribbean being excluded from these spaces [105]. This exclusion risks skewing AI global conversations towards the interests of a few powerful geopolitical and private sector actors in the most affluent countries [36]. As Chowdhury [106] suggests, the participation gap cannot be improved without addressing the “limited technical expertise, staffing, and funding” that puts global majority policy and governance actors under a more significant burden and alienation.

Because of the relentless push by powerful companies and political unwillingness to restrain them and the complex, transnational, and multifarious entanglement of these technologies, AI appears ungovernable. These global challenges reflect the status quo of AI which reinforces a narrative of dominance by AI superpowers over the future of society and the planet. In the rest of the paper we aim to confront these challenges through a deeper engagement with the spatial, temporal and political dimensions of AI, ending with proposing an actionable agenda for just and sustainable transformation.

3 Infrastructural inversion

In this section we introduce an analytical and methodological approach known as “infrastructural inversion” which allows us to shed light on the largely hidden properties, relations, and accountabilities of the AI stack. We use this approach, not only to advance a richer understanding of the technology, but as a means to identify potential bottlenecks, sites of power concentration, and relatedly opportunities for strategic governance intervention which may lie beyond the default focus on governing the affordances and capabilities of AI.

AI technologies have persistently been described as “black boxes” [107–109]. Technical experts and AI researchers often refer to the “unexpected” and “emerging”

capabilities of technologies like foundation models—a stark reminder that even technical experts admit to not always understanding what is going on [91]. Many researchers in the field of AI have had a lasting commitment to advancing explainability and different ways of translating and making complex technical processes more intelligible and transparent [110–112]. But technical transparency alone does not automatically lead to better understanding and accountability. As Ananny and Crawford observe, “Not only is transparency a limited way of knowing systems, but it cannot be used to explain—much less govern—a distributed set of human and non-human actors whose significance lies not internally but relationally” [113].

Indeed, more attention tends to be given in public fora to understanding functionality and technical concepts than to the relations between actors, practices, and norms that shape AI production and use. The sociotechnical assemblages of AI—that is, the compute, data infrastructures, labour required to clean the data used to train the models, and the material resources needed for the operation—remains largely removed from public scrutiny. This persistent opacity is a challenge to advancing public understanding, trustworthiness, and effective governance. Understanding the ecology of actors and their interdependencies is critical to the design of effective safeguards, policies, and mechanisms of accountability.

Having said that, the understanding of AI not merely as algorithms or software, but as assemblages of people and technology—or as sociotechnical phenomena—is well established in academic circles and has been increasingly mobilised to advance policy-oriented knowledge of artificial intelligence (see e.g. [34, 114]). Expanding on this understanding, we argue that in order for governance interventions to properly address the systemic and planetary challenges posed by AI, they must reckon with the technology’s globally distributed layers and relations through a sociotechnical lens. This entails expanding the way AI is conceptualised to account for the social, economic, and political interdependencies arising from the increased space-temporal scale and complexity of this technology.

Approaches to representing the distributed and iterative processes involved in AI have generally leaned on the notion of “technology lifecycle.” Technical reports accompanying the release of large-scale models such as GPT-4 [115] or PaLM 2 [116], for instance, broadly follow the project lifecycle outlined by Bommasani et al., [5] of *data creation*, *data curation*, *training*, *adaptation*, and *deployment*. Governance oriented documents have similarly followed the AI system lifecycle albeit more often in sociotechnical terms integrating considerations of “people and planet” and “economic context” alongside data, model, and deployment components. However, the technology lifecycle on its own

offers a limited, AI-centric view whereby important path dependencies and historical continuities relevant for governance may be overlooked. One cannot ignore, for instance, the role of established technology companies in shaping the AI market, the cumulative learning of surveillance and attention-driven business models of the last two decades, the solidification of a planetary labour market crucial for the data work powering AI, and the vast pool of user-created data on the internet upon which foundation models are trained.

Moreover, one should look beyond the AI lifecycle to reckon with the (geo)political and financial forces shaping AI. In the last few years, an emerging body of literature has been concerned with charting the global supply chains [15, 70, 117, 118] and value chains [119, 120] of AI, drawing out key aspects of the technology’s political economy, its sociomaterial entanglements, and broader ethical implications [58, 121, 122]. In this paper, we bring together different strands of this work to capture the wider ecology of actors, practices, and artifacts constituting contemporary AI systems and its interplay with AI governance.

We take inspiration from information infrastructures scholarship within the field of STS, which offers a helpful analytical lens for surfacing the variously hidden properties of large technological systems and their constitutive social relations. Information infrastructures scholars have argued that as technologies become more and more mainstream and people come to rely upon them, the hardware, labour, conventions of practice, standards, and other processes that support them tend to become relatively invisible and taken for granted [123, 124]. The level of awareness we have about a technology’s innerworkings will largely depend on a range of contextual factors, which may be shaped by our interests, expertise, or relative proximity to infrastructural processes and practices, to name a few. In other words, the visibility of large systems is a relational and variegated property depending on one’s vantage point and positionality [125]. For instance, consider the internet where engineers and other IT practitioners may be more closely attuned with less visible technical layers (e.g., submarine cables or internet protocols) compared to everyday internet users. Because of this, the infrastructure and various types of “practices, uses, and exchanges” underpinning digital technologies tend to remain out of sight for many, only becoming more salient in moments of failure [126, 127]. This is acutely true in the case of large scale AI models where key relations such as the critical human work of data curation and labelling necessary to make systems safer can easily be overlooked in conversations about balancing AI risks and opportunities. Concerns and calls for action have in fact been brought to light by data workers and researchers advocating safer and less precarious working conditions or local communities

directly impacted by the water scarcity caused by large data centres [70].

The term “infrastructural inversion” was coined by Bowker [128] and later expanded upon in Bowker and Star [124], to refer to the deliberate act of “going backstage” and turning attention to the less visible aspects of sociotechnical assemblages. This approach can not only be used to advance understanding of the interdependencies of complex technical systems, but as an empirical foothold to inform and situate governance interventions be they technical, organisational, legal, or market based. Due to the globally distributed nature of these systems, understanding specific sites of technical and non-technical intervention and accountabilities can be crucial to the design of effective governance.

The lens of infrastructural inversion reminds us that AI systems do not arise *de novo* but are situated in a pre-existing world, making AI both a product and a driving force of multifarious social transformations. By representing AI systems in this way, we aim to make the technology more publicly intelligible and contestable, challenging grandiose narratives that frame new technologies in mystical terms or that portray them as inevitable, magical, or ahistorical phenomena. We aim instead to understand AI as socially shaped, governable artifacts whose uses can be restrained or directed to deliberate goals such as protecting the planet, human rights, and collective wellbeing.

4 Toward a sociotechnical ecology of AI

In this section, we unpack the sociotechnical arrangements that make up the AI ecology identifying the prevalent actors and their mutually shaping relations. In the foundational work of infrastructure studies, the ecological imaginary is used to represent interdependent, potentially symbiotic relations between collaborating entities as well as circumstances and factors leading to unbalance [129]. The term “ecology,” according to Star and Ruhleder [129], “refers to the delicate balance of language and practice across communities and parts of organisations; it draws attention to that balance (or lack of it). It is not meant to imply either a biological approach or a closed, functional systemic one.” With this frame, we aim to map out the key actors and practices of AI production, dissemination, and integration at each of various layers of the technology. Through this mapping, we also locate the key sites of decision-making contributing to the global AI governance challenges, highlighted in Sect. 2.

We ground our analysis on the existing technical and sociotechnical literature, with the aim to trace the key ingredients and interdependencies that make AI systems possible [130–132]. Drawing from existing conceptual

representations of AI in the literature [122, 133, 134] we explore the following four layers as constituting the AI stack:

- **Compute infrastructure:** Comprises the material components, energy sources, real estate, human activities, and financial investment involved in AI production.
- **Data pipelines:** Comprises the network of human labour and vast data sets created, curated and synthesised to form the training and testing data for AI models.
- **Algorithm and model production:** Entails a series of interlinked practices and actors involved in architecting and developing AI models. Depending on the complexity of the models, these may include model selection and training, model pre-training, benchmarking and evaluation, adaptation and agentification, among other practices.
- **Application development and deployment:** Entails the packaging of AI models into downstream applications and their integration and distribution into platforms and other systems through the use of application programming interfaces (APIs) and user interfaces.

Recognising that AI is a rapidly evolving technology, this representation is necessarily tentative and non-exhaustive, serving primarily as a conceptual scaffold to make visible potential sites of accountability, power, and agency, and address some of the challenges of global AI governance. We map three aspects in the sociotechnical ecology. First, the actors and practices involved in AI production; second, their interdependencies; and third, how these relationships have been shaped over time by historical precedents and path dependencies. We call these *infrastructural continuities*.

4.1 Actors and practices in AI production

The evolving ecology of AI innovation comprises an array of interdependent actors who influence and are influenced by the production, dissemination, and integration of AI in society. In Table 1 we draw on existing literature [39, 41, 45, 51, 52, 70, 120, 130, 133] to identify a set of key (primary and secondary) actors operating at each of the layers of the AI stack, their practices and their relevance to global governance challenges.

We represent primary actors as those having direct control and oversight over constitutive components or functions of AI production. These include for example, hardware providers like Nvidia who provide graphic processing units (GPUs) critical for pretraining foundation models, big technology companies and well funded AI labs involved in research and development, the data brokers and intermediaries procuring the necessary inputs for AI models, and the

Table 1 Key actors, practices, and associated global governance challenges across the AI stack

Layer	Primary actors operating in the layer	Practices of key actors	Secondary actors operating in the layer	Associated AI governance challenges
Compute infrastructure	A range of firms are responsible for sourcing the computing and other material requirements: chip designers and manufacturers (e.g., Nvidia, TSMC), datacenter providers/operators, cloud infrastructure providers (e.g., AWS, Azure, Alibaba), data center real estate companies, mining conglomerates.	Design and manufacturing of chips, Provision of datacenters, and cloud infrastructure services.	Compute infrastructure is entangled with mining conglomerates that extract the resources needed for the hardware, including silicon, copper, tantalum, aluminium, and tungsten.	Infrastructure concentration and geopolitical competition for compute supremacy; Environmental degradation and human rights violations (in conflict minerals zones);
Data pipelines	Data providers, data brokers, AI data work intermediaries (e.g., microwork and crowdsourcing platforms, business process outsourcing firms like Sama and crowdsourcing platforms like Amazon Mechanical Turk).	Collecting, curating, annotating, and providing/ hosting the datasets that train and refine AI models.	IP and data regulators, data workers, data rights holders (e.g., copyright holders, licensors, data subjects).	Threats to digital rights due to personal data extraction, non-consensual scraping of extraction of creative works, data work exploitation.
Algorithm and model production	Big tech firms and AI labs producing foundation models and applications. Core AI model producers include OpenAI, Google Deepmind, Microsoft, Anthropic, DeepSeek, Alibaba.	Research and development. Model producers employ diverse release strategies that can be placed across a spectrum of openness in terms of access to the model, components facilitating evaluation and model replication.	Digital market and competition and antitrust regulators, auditors, AI Safety/Security Institutes, standards bodies, and investors.	Power concentration due to market consolidation around a few big players;. Self-preferencing, mergers and acquisitions, and strategic alliances gatekeeping competition. Homogenisation of outputs and epistemic harms.
Application development and deployment	Downstream developers, open model hubs and hosting services (e.g., HuggingFace, GitHub), and distribution platforms (e.g., Apple App Store, OpenAI GTMP Store).	Firms and innovators adapt foundation models into user-facing applications and decision-making systems. Model hubs host and make available open-source AI models.	Stratified layers of AI end-users and consumers (e.g., military, corporate, public sector, individual); Sectoral regulators; Third-party auditors; Investors.	Downstream harms (e.g., cybersecurity, discrimination, mis/disinformation, toxic content)

downstream developers of AI applications. We also identify secondary actors who play an important albeit indirect role in the AI ecology by either contributing with data (as data subjects and rights holders), venture capital and finance, normative insight, evaluations, or regulatory oversight of the technology. We locate some of the most critical governance challenges at each of the layers of the stack drawing on the body of existing literature explored in Sect. 2.

4.2 Interdependencies

The path and pace of AI production are contingent on material resources, energy, finance, market dynamics, data availability, labour relations, and pressures from regulators, academia, and civil society. Actors who shape and control components of AI production, dissemination, and integration are mediated by cross-border flows of capital, data and information, energy and natural resource markets, labour

markets, and legal and regulatory frameworks. Not only do they enable the practices of other actors across the AI value chain, but their own practices depend on both downstream and upstream activities. Far from a linear innovation process, AI production is inflected by feedback loops and moments of iteration. As AI models integrate into society, continuous feedback and refinement takes place as user interactions become observable to AI engineers, developers and policy makers. This provides opportunities for altering both the models and the interfaces and introduce guardrails to align them with users' needs and other objectives. Feedback loops are important governance entry points not only for spotting areas of refinement and enhancing user experience, but for addressing societal and ethical considerations including privacy protection, intellectual property rights, and the risk of manipulation, amongst other issues of responsible production and use.

As noted by Cobbe et al. [15], these interdependencies are uneven (some actors hold more influence than others), unstable, and “structured by technological, legal, and political economic factors”. Because of their dynamic nature, we do not intend to exhaustively map interdependencies, but we explore some of them here.

At the compute infrastructure layer, Nvidia dominates chip design and outsources manufacturing to other companies, mainly the Taiwan Semiconductor Manufacturing Company (TSMC) [70]. Some technology companies, such as Microsoft and OpenAI, however, are now developing their own [135, 136]. Chip manufacturing is strongly reliant on outsourced mining companies, who supply the silicon, copper, tantalum, aluminum, tungsten, cobalt, tin, and other materials required to make GPUs to train AI models. Mining companies privilege locations where not only is extraction cheaper but also less regulated, despite human rights violations and risks to microbiological diversity and local sustainability. Such is the case of the Democratic Republic of Congo's exacerbation of violent conflict over mineral extraction [137] and the Atacama Desert in Chile, where lithium “is extracted by pumping brine pools that are left to evaporate on the surface” [138].

The production of large scale AI models heavily depends on data work to train, test, quality-check, and update AI models. Rather than conducting time and labour-intensive processes of data cleaning and labelling in-house, core AI producers, such as OpenAI and Meta, often outsource these tasks to business processes outsourcing (BPO) companies, many of which operate in the Global South. These practices are motivated by the profitability and commercial advantages of leveraging a globally connected workforce and exploiting the political economic conditions of the jurisdictions where delivery centres are based [52]. BPO companies rely on low-paid labour in the Global South to fulfill project

requirements for clients in wealthier countries. The BPO industry, often self-described as “impact sourcing,” further profits from socially and economically disadvantaged workers by offering short-term and intermittent work [139]. BPO firms also take advantage of regulatory and accountability gaps in the locales where they operate due for example to more less stringent data and labour rights [139]. Sama, an impact sourcing BPO firm based in San Francisco has been criticised for the low wages offered to data workers in its delivery centres in East Africa whilst signing contracts that would serve some of the most expensive AI projects by big tech clients. For instance, the company paid its data workers around \$1.32 and \$2 per hour for curating datasets for OpenAI's ChatGPT and \$170 per month to the most junior workers in Kenya for another OpenAI project that entailed labelling content with references to several types of abuses and violence [140].

Current market incentives have encouraged vertical integration and the formation of technology conglomerates, some of which have larger market valuations than the GDP of entire states [141]. Well-funded companies have an incentive to use their dominant position to engage in strategic mergers, acquisitions, and partnerships aimed at gaining exclusive access to talent, large datasets and other infrastructural assets to improve and scale up AI models. Some recent examples include Microsoft-owned LinkedIn and Meta decisions to use personal data from their social media platforms to improve GenAI models [142, 143], and a series of recent deals ensuring Big Tech companies access to large datasets from academic publishers and news organisations [144].

Likewise, some companies have engaged in partnerships to enhance the AI features of their products and gain competitive advantage whilst also providing AI creators access to a new user base. A notable example is the partnership between Apple and OpenAI, which enabled the integration of ChatGPT into Apple's operation systems. With user consent, this integration also allows ChatGPT to assist Siri with some user requests [145]. By integrating GenAI capabilities into its own AI models, Apple offers enhanced functionalities to both Apple and third-party applications. This strengthens the role of Apple in the AI market while further expanding the distribution of OpenAI models. A similar deal has been struck between Apple and Alibaba to serve the Chinese market [146].

4.3 Infrastructural continuities

It is crucial to examine AI not merely as a layered technology stack, but a sociotechnical assemblage built upon pre-existing infrastructures and within environmental, social, cultural, political, and economic contexts. Contrary to

mainstream corporate imaginaries that frame new AI models and systems as unprecedented, we view them as shaped by deep path dependencies and longitudinal patterns. Beyond the actors and their relationships, we identify what we call *infrastructural continuities*, with the aim to make explicit how AI leverages and amplifies existing patterns of asymmetry and dependency.

Pre-existing IT and data infrastructures: Current large AI models benefit from existing built infrastructures (databases, protocols, electricity, water systems, real estate, etc.) and crucially, the internet, not only for their distribution but for their training. Datasets used to train large models are generally drawn from massive repositories of open and proprietary data. These datasets are primarily generated from human interaction with online systems, which may include any user-generated content and metadata available and accessible on the internet including, among others, Wikipedia entries, social media posts, virtual meeting participation, uploaded videos, search queries, and different types of knowledge databases including books, journal articles, and other sources. Importantly, the outputs of AI systems are also altering the fabric of the internet with synthetic data which in turn becomes available as training data. This is likely to lead to future generations of AI systems being trained largely on synthetic content, which could threaten their viability [147].

Planetary labour market: A largely hidden aspect of AI is the contribution of data workers in the curation, cleaning, and labelling of data used for training. Global labour supply chains are not novel arrangements: they are established in domains like customer service, hardware manufacturing, and textiles [148]. The planetary labour market has evolved in the last two decades alongside the global increase in internet access leading to the proliferation of outsourcing and offshoring of data and digital piecework, but also of data science and software engineering work, to countries in the Global South with lower wages and labour protections [52, 149]. In the era of GenAI, the conditions of AI data work have become a pressing governance issue in the AI data supply chain with increasing calls for improving transparency, fairness, and worker wellbeing, given the precarious and psychologically unsafe nature of much of this work [51].

Existing market power of technology companies: The current AI market is dominated by a limited number of companies, including newly formed AI labs as well as established technology companies with the resources and market power to finance AI research and development. Established players like Microsoft, Google, Meta, and Nvidia benefit from the long-term network effects provided by their existing user bases and accrued technological lock-in. This entrenched power is strategically leveraged to propel market capture,

acquire and gatekeep competitors, lobby policymakers and regulators, and mitigate financial losses [38].

Platformised business models: The dominant paradigm of AI commercialisation follows the now mature platform business model consolidated by large technology companies in the last two decades [150]. For instance, Large Language Models (LLMs) produced by firms like OpenAI are made accessible to a distributed array of actors who evaluate, integrate, adapt, and use them through APIs. The platformised business model puts dominant firms in a core position that allows them to offset regulatory, financial, and ethical risks to smaller firms while centralising value capture [151].

These long-term continuities underscore how AI adapts through, rather than breaks with, historical patterns and relations. Governance interventions must confront these structural path dependencies to redirect innovation toward more sustainable and equitable outcomes.

5 Four governance imperatives for addressing power asymmetries in the ecology of AI

The ecological perspective surfaces symbiotic relationships between some actors but also a host of failures for others at the receiving end of societal and environmental harms. Two mutually reinforcing factors leading to power concentration are the entrenched market position of Big Tech companies and the large-scale, data-intensive architecture of large AI models that reinforce an oligopolistic centralisation of data, compute infrastructure, and talent.

While these factors can easily be glossed over as the inevitable status quo or posing too complex or abstract challenges, our ecological lens lays bare multiple layers and moments of accountability and potential intervention. We view these as concrete opportunities for confronting the systemic challenges outlined at the beginning of this paper.

Importantly, recent scholarship and international organisations have intensified efforts to design more systemic interventions. Such initiatives aim to recalibrate incentive structures and address power asymmetries in the political economy of AI. Indeed, global AI governance has fallen squarely within the purview of transnational policy actors like the United Nations, the Group of 20 (G20), UNESCO, and the OECD. These actors have been actively mobilising, albeit with varying degrees of success, international policy, standards, cooperation, and harmonisation of regulation through initiatives like the Global Partnership on AI (GPAI) and more recently the formation of a UN Independent International Scientific Panel on AI. However, there is a role for more localised regional efforts, particularly in the Global South, aimed at locating levers of action across the different

layers of the AI stack and counterbalancing the geopolitical hegemony of the “AI superpowers.” These efforts foreground governance paradigms that prioritise equitable resource distribution, epistemic diversity, and decentralised innovation strategies over solely market-driven logics.

In the following we identify four key imperatives for reorienting global AI governance aimed at challenging the current status quo of power concentration in AI production and governance. We acknowledge that there are no silver bullets to address the complex global challenges posed by AI. The following imperatives should therefore not be interpreted as offering 1-to-1 mappings to the challenges outlined in Sect. 2, but as necessarily overlapping, complementary, and mutually contingent. As we will illustrate, one governance imperative could speak to more than one systemic challenge, and conversely, a systemic challenge may be addressed by more than one imperative.

5.1 Decentralising AI infrastructure

Controlling key AI infrastructure offers a direct form of actionable governance which could be directed to both power concentration and distribution. In the absence of enforceable regulatory mechanisms beyond national or regional borders, infrastructure control has always been a concrete leverage by both private and public actors to govern flows of data, capital, and labour. This phenomenon is not new and has been described by internet governance scholars as “governance by infrastructure” [126].

Digital Public Infrastructures (DPIs) and AI sovereignty frameworks have gained traction as potential countermeasures to mitigate technological dependence on powerful actors in the AI ecology. The concept of DPIs emphasises the role of the state in exercising control over critical infrastructure to serve the public interest, improve access to digital services, or to advance strategic goals of national security and economic sovereignty [152, 153]. Similarly, the discourse of AI sovereignty has been proposed as a way forward at multiple levels while simultaneously raising tensions between competing interests—for instance, between geopolitical competitiveness and security and domestic socio-economic goals. At the EU level, the concept has been driven by aspirational goals of aligning AI innovation to European values and controlling key infrastructural elements within jurisdictional limits [154]. In some Global South countries like in the case of Brazil, the discourse of AI sovereignty has been more closely associated with increasing agency and self-determination and overcoming dependence on foreign AI systems [155]. And at the community level, the notion of sovereignty with regards to data practices has long been pursued by civil society [156] and indigenous

communities such as Māori [157] and has recently gained prominence within the context of AI.

Building regional or locally situated data and compute infrastructure aimed for example at supporting AI research or local production of foundation models or LLMs is a concrete approach to overcoming infrastructure lock-in and dependence. However, we caution that such efforts must be oriented to advancing democratic aims rather than helping countries to compete in the global AI race. The latter risks a potential replication of Big Tech strategies by the state, particularly when it comes to surveillance and unsustainable scaling. Indeed, much of the sovereignty discourse is driven by national security and geopolitical races, which has been criticised as undermining cooperation for global AI governance [158].

We argue that infrastructure decentralisation should be viewed not only a measure for mitigating technological dependence on a few companies but as a mechanism to advance more sustainable and equitable innovation globally. While the case of DeepSeek can be regarded as offering a cost-efficient alternative that decenters Silicon Valley AI models, merely shifting from an AI superpower to another does not remove technological dependence faced by the rest of the world. This underscores the need to frame infrastructure decentralisation goals explicitly as a matter of collaboration, public interest, and sustainability—anchored in poly-centricity rather than oligo-centricity. Global south scholars have called to reclaim digital sovereignty through decolonial lenses, advocating for governance models that resist corporate extractivism and empower local economic development through ethics and politics that favor human interdependence, autonomy, and self-determination [159–161]. Rikap et al. [160] propose to “ground digital sovereignty in an ecological internationalism that rejects seeing sovereignty as a battlefield among countries, which neglects that today’s rulers are not only powerful states but also leading corporations”.

Decentralisation initiatives like the Distributed AI Research (DAIR) Institute, or the notion of Public AI not only advocate for increased public interest AI systems and applications but emphasise the need to decentralise the entirety of the AI ecology with a focus on avoiding Big Tech monopolies, ensuring public access, public accountability, and permanent public goods [162]. Proponents of decentralisation underline how this endeavour cannot be spearheaded by a single actor or entity but rather requires concerted action from everyone including civil society, developers, policymakers, academics, companies, and the public [163].

5.2 Advancing environmental and epistemic justice through pluriversal AI governance

Environmental concerns have been centered in critiques about the scale and scope of the current dominant paradigm AI production. To that end, a growing repertoire of public control mechanisms have been proposed including the implementation of ex-ante environmental impact assessments [164]; methods for measuring, tracking, and reporting environmental indicators for AI-intensive processes [see e.g., 63], and democratic oversight enabling affected communities to contest projects threatening local ecosystems [70]. Crucially, these measures must take into account wider sociohistorical concerns of human rights within data infrastructures and systems [165–167].

Reaching global consensus in enacting comprehensive governance measures that seek to address and mitigate the adverse environmental impacts of AI has arisen as one key international governance goal. Yet concurrently, critical scholars have challenged potential universalist sustainability paradigms arguing for more situated governance grounded in pluriversal epistemologies that recognise diverse relational ontologies between technology, humans, and the planet [168–171]. The concept of the pluriverse as framed by Arturo Escobar [169], is a recognition of the coexistence of many Worlds but also of “plural ways for making the World”.

The representation of environmental and ecological knowledge in AI systems and the translation of environmental evidence by LLMs, such as ChatGPT, suggest glaring biases but also the implicit reinforcement of knowledge hierarchies [172]. Recent studies point to biases in environmental knowledge production and complex literacies, including global conservation injustices in research and practice, in these systems which remain dominated by Western knowledge, with lack of diverse representation of sources of expertise and perspectives across Global South countries and local traditional knowledge holders [172, 173].

Universalising solutions to complex social and planetary challenges like those offered by large scale AI technologies raise a deeper, less visible, issue of epistemic injustice. As foundation models trained on Western knowledge and value systems get distributed globally, they impose these values onto non-Western contexts. This not only risks causing harms related to inaccuracies and biases, but a devaluing, subordination and potential erasure of alternative ways of knowing, living, and addressing social challenges [174]. As noted by Bennet et al. [175], exports of LLMs impose a hierarchy of knowledge through the use of hegemonic languages whereby the issue of homogenisation is framed in technical terms is addressed accordingly through technical fixes like finetuning.

Decolonising LLMs requires more concerted efforts in practice and discourse if the concurrent social, political and environmental crises are to be addressed—safeguarding, in particular, those most vulnerable human and ecological systems. Encapsulated in pluriversal AI governance is the notion of contextualised, culturally relevant, and ecologically sustainable innovation approaches that uphold epistemic plurality and transitions to more just futures [173, 176, 177]. This reflects a recognition that other legitimate forms of knowing exist beyond Western epistemology. Indeed, indigenous and historically marginalised communities have long offered pathways to planetary flourishing grounded in ancestral knowledges, relationality and bio-centric ethics. Decisions surrounding whether and how AI systems are designed and used should always be responsive to local ecological, cultural, and political contexts where AI infrastructure has a direct impact. Locally responsive governance for instance must reckon with the impacts of building data centres on local communities who will be compelled to protect their livelihoods, natural resources, territories, and cultures [70].

Beyond oversight of the ecological footprints of large AI producers, and policies aimed at improving efficiencies, a concrete response to the high social, epistemic and ecological risks of large-scale AI is found in the pursuit of paths to AI innovation that prioritise more focused, localised, and small-scale systems. We can draw inspiration from existing initiatives advancing local AI innovation in the Global South. Biyela et al. [63] provide an in-depth analysis, describing more than 10 regionally developed non-English LLMs across six countries (India, South Africa, Kenya, Saudi Arabia, UAE, and Qatar), targeting languages including Swahili, Hausa, Yoruba, Zulu, isiXhosa, Arabic, and several Indian languages. Initiatives like Chile’s Latam GPT, a project to develop a Latin American Language Model [178], InkubaLM, a Small Language Model for five widely spoken African languages [179], and monolingual MariTalk developed in Portuguese by Brazil’s Maritaca AI [180], are exemplars of how local actors are leveraging open-source frameworks and pooling infrastructural resources to assert digital sovereignty and develop more localised and culturally relevant models.

5.3 Instituting cross-border data and data work governance

Data governance is an essential consideration in collective AI governance [181]. National and regional data protection laws are viewed as a distinct lever which could be (re) aligned to AI governance [182]. Existing measures incorporate specific frameworks on data usage, training data accountability, anonymisation standards, among others

[183]. However there exists a gap when it comes to cross-jurisdiction protection of fundamental rights like privacy, dignity, and non-discrimination, and ensuring fair use of public data and open data repositories by AI companies. Cross-border data governance can be advanced through the establishment of international and regional bodies entrusted to ensure accountability and transparency of data provenance [184, 185]. Promising data governance models that facilitate cross-border data sharing whilst protecting rights and interests are steadily emerging, such as data commons, data foundations, data cooperatives, and trusts [186, 187]. These models can support distributed or public ownership within a legal framework which allow individuals and communities to maintain control over how data are used [187] and which can protect data workers and rights holders and/or involve them in the co-creation of data governance measures [188, 189].

Data workers around the world, who make up a key link in global data supply chains, lack basic protections from exploitative and psychologically harmful working conditions [190]. There is a need for cross-border transparency and enforceable provisions to safeguard data workers and realise more just data supply chains. For instance, through due diligence legislations—and accompanying standards and guidelines for implementation—that require technology companies to audit their providers for compliance with labour and human rights, including those activities taking place outside their jurisdiction [148, 191].

Some scholars suggest establishing infrastructures for supporting data worker organisations and unionisation [51, 191] and rethinking the relationship between data workers and AI production [192]. There is a need for a paradigm shift aimed at making data work visible and improving the working conditions and agency of data workers. International efforts could contribute to the development of frameworks that promote the engagement of data worker organisations and unions in decision-making within AI governance processes [193].

International multi-stakeholder data governance is positioned and expected to facilitate partnerships with a wider range of stakeholders (e.g., industry, regulators, civic organisations and academia), such as the Global Partnership on Artificial Intelligence (GPAI) and a UN international panel on AI with a focus on data work which can further reinforce regulatory measures across global data supply chains [105]. Given the power asymmetries across AI production, it is crucial that international bodies and panels do not replicate uneven patterns of representation and/or decision-making in the pursuit of multi-stakeholder and collective data governance [105, 194].

5.4 Enhancing international coordination, participation and solidarity

Enhanced coordination around nation-level capacity-building and governance within country states and across consortia of states is imperative to sharing infrastructural resources in a sustainable way and improving the bargaining and enforcement power of regulatory strategies. This is more crucial in the Global South where regional alliances can help to countervail the power of large technology firms through resource pooling and coordinated action, as opposed to environmentally costly AI races.

In limiting the concentration of power by large firms, governments also need to avoid endorsing the extractive financial models of large technology firms and focus instead on incentivising competition through industrial policy and responsible public procurement strategies [194]. This may involve restricting the lobbying of dominant companies on regulatory discourse and decision-making and incentivising the participation of smaller actors and open-source systems in the AI ecology.

The interoperability of regulatory measures is required to assign ownership and accountability across the AI value chain. These necessitate cross-jurisdictional cooperation on AI issues such as data privacy, cybersecurity, and ethical AI use; auditing of emissions and water use linked to model training using standardised metrics [195]; and cross-border enforcement of non-discrimination laws during model deployment. Without regional alignment, the structural power asymmetries enabling today's AI oligopolies will persist, foreclosing alternative innovation pathways.

Given the transnational dependencies and historical continuities of AI, it is imperative that the design and implementation of governance measures meaningfully involve public deliberation and international solidarity. Initiating and sustaining models for global participation and solidarity in AI governance is crucial for ensuring that those bearing the brunt of harms and adverse impacts stemming from increased AI adoption are not sidelined [196]. Impacted individuals and communities as well as wider publics should be able to play an important role in deliberating and addressing the critical questions arising within AI governance. Approaches to global citizen deliberation have been explored by Davies et al. [197] who have developed recommendations for an overarching commissioning body for public dialogue as well as four distinct options for assembly: deliberative review, global assembly, distributed dialogue, and collective intelligence. Although each approach carries its own strengths and weaknesses, they all highlight the urgent need for operationalising practicable approaches to sustaining proactive public participation in AI governance.

6 Conclusion

The rapid development and global distribution of AI systems present significant social and planetary challenges arising from their complex sociotechnical ecology. This ecology is shaped by transnational flows of data, labour, and materials, influenced by geopolitical competition and fragmented regulatory environments. The most influential AI governance efforts to date—particularly those framed around AI safety and security—tend to focus on downstream issues, such as bias and misinformation, while overlooking upstream harms and systemic issues related to power concentration, labour exploitation, data extraction, and environmental injustice.

We trace the making of contemporary AI models, as a means to identify concrete sites of decision-making which could signal strategic opportunities for agency, cooperation, solidarity, and accountability. Using the lens of infrastructural inversion, we highlight how AI technologies are embedded in longitudinal structures and relationships. This perspective helps us, both conceptually and empirically, to make more apparent the interplay between key actors implicated in the making of these technologies, as well as the historical continuities and path dependencies shaping current trends.

Through this lens we advance an agenda of transformative action identifying levers of governance that can address the complex social and biospheric challenges posed by AI. We argue that tackling these global challenges requires a multi-pronged and multi-level approach to governance—one that involves international and regional coordination as well as localised action and forms of governance from below. We outline four priority areas of research, civic, and policy action aimed at building more sustainable and equitable futures.

First, we advocate for decentralising infrastructure by envisioning more collective ways to own, manage, situate, and sustainably share key resources. This is crucial for countervailing power concentration and enabling more democratic participation. Second, we call for resisting universal paradigms of AI-driven innovation and AI-driven climate crisis solutions in favour of strengthening institutions and mechanisms that pluralise responses to societal challenges. There is much to learn from the efforts of scholars and civil society in the majority world—voices that have been largely sidelined in the most influential fora on global AI governance. Third, we propose a human rights oriented approach to governing the flows of data, information and labour—an approach underpinned by intergovernmental systems as well as bottom up participation of impacted communities. Finally, there is an urgent need to reorient AI production away from ecologically damaging races toward more sustainable cooperation and international solidarity.

There are numerous inspiring precedents and ongoing initiatives already addressing aspects of these governance imperatives. However much work remains to enact the infrastructures, institutions and supranational mechanisms necessary to meaningfully alter the existing perverse incentives that drive the development and deployment of AI. Transforming these underlying dynamics is critical for reversing patterns of monopolistic extraction and entrenched power. In response to what are often considered complex and seemingly immutable political, economic and ecological crises, we argue for a holistic, collective, decentralised, and pluriversal approach to governing AI—one grounded in the pursuit of social and planetary wellbeing.

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Declarations

Competing interests The authors declare no competing interests.

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